

## 1. Executive Summary

| Metric                  | Value                   | Traffic Summary        | Count      |
|-------------------------|-------------------------|------------------------|------------|
| Report Period           | 2026-08-31 - 2026-09-07 | Total Flows Observed   | 13205      |
| Total Threats Mitigated | 50                      | Total Packets Analyzed | 3586596503 |
| High Priority           | 17                      | Malicious Packets      | 3586057137 |
| Low Priority            | 33                      | True Negatives Passed  | 6951       |
| Manual Releases         | 0                       |                        |            |
| Manual Blocks           | 0                       |                        |            |
|                         |                         |                        |            |
| ML Metrics              | Value                   | Attack Vector          | Count      |
| False Positives         | 10                      | ICMP Flood             | 4          |
| FP Rate                 | 0.14%                   | SYN Flood              | 16         |
| IF Precision            | 99.84%                  | UDP Flood              | 13         |
| IF Recall               | 99.35%                  | Uncertain              | 17         |
| IF F1-Score             | 99.59%                  |                        |            |
| IF Accuracy             | 99.61%                  | Action                 | Count      |
|                         |                         | Quarantined            | 12         |
|                         |                         | Time Ban               | 7          |
|                         |                         | Blackhole              | 0          |
|                         |                         | Released               | 7          |

## 2. Performance Benchmark

### 2a. Isolation Forest-Anomaly Detection

| Metric                    | Value  | Description                                          |
|---------------------------|--------|------------------------------------------------------|
| Precision                 | 99.84% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 99.35% | Share of actual attacks successfully flagged         |
| F1-Score                  | 99.59% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 99.61% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.14%  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 0.65%  | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 99.35% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 99.86% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 6244           | FN 41             |
| Actual: Normal | FP 10             | TN 6910           |

### 2b. Random Forest-Attack Classification

| Metric    | Value   | Description                                                  |
|-----------|---------|--------------------------------------------------------------|
| Precision | 100.00% | Share of classified flows assigned the correct attack type   |
| Recall    | 100.00% | Share of attacks of each type that were correctly identified |
| F1-Score  | 100.00% | Balanced measure combining Precision and Recall              |
| Accuracy  | 100.00% | Overall proportion of correct classifications                |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 4574           | 0               | 0              |
| Act: ICMP | 0              | 251             | 0              |
| Act: UDP  | 0              | 0               | 1417           |

### 2c. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 729.1 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 6.7 ms   | Interval between the anomaly flag and the FlowMod blocking rule install |

2d. Controller Resource Overhead

| Metric                         | Value | Description                                                           |
|--------------------------------|-------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 1.12% | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 7.89% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 7.89% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.16 | 3        | 3            | 2026-09-07 14:00:37 | 2026-09-07 14:03:43 |
| 10.0.0.17 | 3        | 3            | 2026-09-07 14:00:09 | 2026-09-07 14:03:15 |
| 10.0.0.18 | 3        | 3            | 2026-09-07 14:00:09 | 2026-09-07 14:03:15 |
| 10.0.0.19 | 3        | 3            | 2026-09-07 14:00:09 | 2026-09-07 14:03:15 |
| 10.0.0.21 | 3        | 3            | 2026-09-07 14:00:10 | 2026-09-07 14:03:32 |
| 10.0.0.22 | 3        | 3            | 2026-09-07 14:00:27 | 2026-09-07 14:03:42 |
| 10.0.0.25 | 3        | 3            | 2026-09-07 14:00:13 | 2026-09-07 14:03:26 |
| 10.0.0.8  | 3        | 3            | 2026-09-07 14:01:05 | 2026-09-07 14:03:38 |
| 10.0.0.6  | 3        | 1            | 2026-09-07 14:01:02 | 2026-09-07 14:03:26 |
| 10.0.0.20 | 2        | 1            | 2026-09-07 14:00:16 | 2026-09-07 14:00:56 |
| 10.0.0.23 | 1        | 1            | 2026-09-07 14:00:20 | 2026-09-07 14:00:31 |
| 10.0.0.24 | 1        | 1            | 2026-09-07 14:00:26 | 2026-09-07 14:00:36 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector     | Confidence | Priority | Action        |
|---------------------|-----------|-------|------------|------------|----------|---------------|
| 2026-09-07 14:00:09 | 10.0.0.17 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-07 14:00:09 | 10.0.0.18 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-07 14:00:09 | 10.0.0.19 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-07 14:00:10 | 10.0.0.21 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-07 14:00:13 | 10.0.0.25 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-07 14:00:14 | 10.0.0.17 | DDoS  | SYN Flood  | 92.9%      | Critical | Time Ban (1m) |
| 2026-09-07 14:00:14 | 10.0.0.18 | DDoS  | SYN Flood  | 91.2%      | Critical | Time Ban (1m) |
| 2026-09-07 14:00:14 | 10.0.0.19 | DDoS  | SYN Flood  | 91.2%      | Critical | Time Ban (1m) |
| 2026-09-07 14:00:14 | 10.0.0.18 | DDoS  | SYN Flood  | 91.2%      | Critical | Time Ban      |
| 2026-09-07 14:00:14 | 10.0.0.17 | DDoS  | SYN Flood  | 92.9%      | Critical | Time Ban      |
| 2026-09-07 14:00:15 | 10.0.0.19 | DDoS  | SYN Flood  | 91.2%      | Critical | Time Ban      |
| 2026-09-07 14:00:16 | 10.0.0.20 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-07 14:00:20 | 10.0.0.23 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-07 14:00:23 | 10.0.0.25 | DDoS  | Uncertain  | -          | Low      | Released      |
| 2026-09-07 14:00:26 | 10.0.0.25 | DDoS  | ICMP Flood | 90.0%      | High     | Time Ban (1m) |
| 2026-09-07 14:00:26 | 10.0.0.25 | DDoS  | ICMP Flood | 90.0%      | High     | Time Ban      |
| 2026-09-07 14:00:26 | 10.0.0.24 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-07 14:00:27 | 10.0.0.20 | DDoS  | Uncertain  | -          | Low      | Released      |
| 2026-09-07 14:00:27 | 10.0.0.22 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-07 14:00:31 | 10.0.0.21 | DDoS  | UDP Flood  | 78.7%      | High     | Time Ban (1m) |
| 2026-09-07 14:00:31 | 10.0.0.23 | DDoS  | Uncertain  | -          | Low      | Released      |
| 2026-09-07 14:00:33 | 10.0.0.21 | DDoS  | UDP Flood  | 78.7%      | High     | Time Ban      |
| 2026-09-07 14:00:36 | 10.0.0.24 | DDoS  | Uncertain  | -          | Low      | Released      |
| 2026-09-07 14:00:37 | 10.0.0.16 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-07 14:00:37 | 10.0.0.22 | DDoS  | Uncertain  | -          | Low      | Released      |
| 2026-09-07 14:00:41 | 10.0.0.22 | DDoS  | UDP Flood  | 72.0%      | High     | Time Ban (1m) |
| 2026-09-07 14:00:41 | 10.0.0.22 | DDoS  | UDP Flood  | 72.0%      | High     | Time Ban      |
| 2026-09-07 14:00:43 | 10.0.0.16 | DDoS  | SYN Flood  | 90.1%      | Critical | Time Ban (1m) |
| 2026-09-07 14:00:44 | 10.0.0.16 | DDoS  | SYN Flood  | 90.1%      | Critical | Time Ban      |
| 2026-09-07 14:00:56 | 10.0.0.20 | DDoS  | Uncertain  | 49.0%      | Low      | Sinkhole      |
| 2026-09-07 14:01:05 | 10.0.0.8  | DDoS  | UDP Flood  | 85.9%      | High     | Time Ban (1m) |
| 2026-09-07 14:01:13 | 10.0.0.6  | DDoS  | Uncertain  | 68.7%      | Low      | Released      |
| 2026-09-07 14:01:15 | 10.0.0.17 | DDoS  | SYN Flood  | 92.9%      | Critical | Time Ban (2m) |
| 2026-09-07 14:01:15 | 10.0.0.18 | DDoS  | SYN Flood  | 92.5%      | Critical | Time Ban (2m) |
| 2026-09-07 14:01:15 | 10.0.0.19 | DDoS  | SYN Flood  | 92.1%      | Critical | Time Ban (2m) |
| 2026-09-07 14:01:26 | 10.0.0.25 | DDoS  | ICMP Flood | 91.9%      | High     | Time Ban (2m) |
| 2026-09-07 14:01:32 | 10.0.0.21 | DDoS  | UDP Flood  | 78.7%      | High     | Time Ban (2m) |

| Timestamp           | Source IP | Class | Vector     | Confidence | Priority | Action        |
|---------------------|-----------|-------|------------|------------|----------|---------------|
| 2026-09-07 14:01:42 | 10.0.0.22 | DDoS  | UDP Flood  | 79.5%      | High     | Time Ban (2m) |
| 2026-09-07 14:01:43 | 10.0.0.16 | DDoS  | SYN Flood  | 91.9%      | Critical | Time Ban (2m) |
| 2026-09-07 14:02:06 | 10.0.0.8  | DDoS  | UDP Flood  | 85.9%      | High     | Released      |
| 2026-09-07 14:02:25 | 10.0.0.6  | DDoS  | UDP Flood  | 90.9%      | High     | Time Ban (1m) |
| 2026-09-07 14:02:32 | 10.0.0.8  | DDoS  | UDP Flood  | 85.9%      | High     | Quarantined   |
| 2026-09-07 14:03:15 | 10.0.0.17 | DDoS  | SYN Flood  | 92.9%      | Critical | Time Ban (5m) |
| 2026-09-07 14:03:15 | 10.0.0.18 | DDoS  | SYN Flood  | 92.5%      | Critical | Time Ban (5m) |
| 2026-09-07 14:03:15 | 10.0.0.19 | DDoS  | SYN Flood  | 92.1%      | Critical | Time Ban (5m) |
| 2026-09-07 14:03:26 | 10.0.0.25 | DDoS  | ICMP Flood | 91.9%      | High     | Time Ban (5m) |
| 2026-09-07 14:03:32 | 10.0.0.21 | DDoS  | UDP Flood  | 78.7%      | High     | Time Ban (5m) |
| 2026-09-07 14:03:42 | 10.0.0.22 | DDoS  | UDP Flood  | 79.5%      | High     | Time Ban (5m) |
| 2026-09-07 14:03:43 | 10.0.0.16 | DDoS  | SYN Flood  | 91.9%      | Critical | Time Ban (5m) |
| 2026-09-07 14:03:57 | 10.0.0.6  | DDoS  | UDP Flood  | 92.0%      | High     | Quarantined   |

## 5. IP Attack History (Completed Sessions)

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.16 | SYN Flood  | 0.6390   | 91.9% | Critical | Phase 2 | 3        | 3m 6s    | 2026-09-07 14:03:43 | Attack persisted - escalating to time ban |
| 10.0.0.22 | UDP Flood  | 0.6126   | 79.5% | High     | Phase 2 | 3        | 3m 1s    | 2026-09-07 14:03:42 | Escalated to Time Ban                     |
| 10.0.0.8  | UDP Flood  | 0.5897   | 85.9% | High     | Phase 1 | 3        | 21s      | 2026-09-07 14:03:38 | Attack Stopped                            |
| 10.0.0.21 | UDP Flood  | 0.6364   | 78.7% | High     | Phase 2 | 3        | 3m 22s   | 2026-09-07 14:03:32 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6393   | 91.9% | High     | Phase 2 | 3        | 3m 0s    | 2026-09-07 14:03:26 | Escalated to Time Ban                     |
| 10.0.0.6  | UDP Flood  | 0.5883   | 90.9% | High     | Phase 2 | 1        | 1m 1s    | 2026-09-07 14:03:26 | Ban Expired                               |
| 10.0.0.17 | SYN Flood  | 0.6317   | 92.9% | Critical | Phase 2 | 3        | 3m 6s    | 2026-09-07 14:03:15 | Attack persisted - escalating to time ban |
| 10.0.0.18 | SYN Flood  | 0.6165   | 92.5% | Critical | Phase 2 | 3        | 3m 6s    | 2026-09-07 14:03:15 | Attack persisted - escalating to time ban |
| 10.0.0.19 | SYN Flood  | 0.6404   | 92.1% | Critical | Phase 2 | 3        | 3m 6s    | 2026-09-07 14:03:15 | Attack persisted - escalating to time ban |
| 10.0.0.8  | UDP Flood  | 0.5915   | 85.9% | High     | Phase 1 | 2        | 21s      | 2026-09-07 14:02:53 | Attack Stopped                            |
| 10.0.0.8  | UDP Flood  | 0.5760   | 85.9% | High     | Phase 2 | 1        | 1m 1s    | 2026-09-07 14:02:06 | Ban Expired                               |
| 10.0.0.6  | Uncertain  | 0.6156   | 68.7% | Low      | Phase 1 | 1        | 11s      | 2026-09-07 14:01:50 | Attack Stopped                            |

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                              |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-----------------------------------------------------|
| 10.0.0.16 | SYN Flood  | 0.6409   | 91.9% | Critical | Phase 2 | 2        | 1m 6s    | 2026-09-07 14:01:43 | Attack persisted - escalating to time ban           |
| 10.0.0.22 | UDP Flood  | 0.6126   | 79.5% | High     | Phase 2 | 2        | 1m 1s    | 2026-09-07 14:01:42 | Escalated to Time Ban                               |
| 10.0.0.21 | UDP Flood  | 0.6398   | 78.7% | High     | Phase 2 | 2        | 1m 22s   | 2026-09-07 14:01:32 | Attack persisted - escalating to time ban           |
| 10.0.0.25 | ICMP Flood | 0.6393   | 91.9% | High     | Phase 2 | 2        | 1m 0s    | 2026-09-07 14:01:26 | Escalated to Time Ban                               |
| 10.0.0.17 | SYN Flood  | 0.6357   | 92.9% | Critical | Phase 2 | 2        | 1m 6s    | 2026-09-07 14:01:15 | Attack persisted - escalating to time ban           |
| 10.0.0.18 | SYN Flood  | 0.6202   | 92.5% | Critical | Phase 2 | 2        | 1m 6s    | 2026-09-07 14:01:15 | Attack persisted - escalating to time ban           |
| 10.0.0.19 | SYN Flood  | 0.6266   | 92.1% | Critical | Phase 2 | 2        | 1m 6s    | 2026-09-07 14:01:15 | Attack persisted - escalating to time ban           |
| 10.0.0.6  | Uncertain  | 0.6106   | 68.7% | Low      | Phase 1 | 1        | 11s      | 2026-09-07 14:01:13 | Attack Stopped                                      |
| 10.0.0.20 | Uncertain  | 0.6435   | 49.0% | Medium   | Phase 1 | 1        | 16s      | 2026-09-07 14:00:56 | Unresolved after quarantine - escalated to sinkhole |
| 10.0.0.16 | SYN Flood  | 0.6332   | 90.1% | Critical | Phase 2 | 1        | 6s       | 2026-09-07 14:00:43 | Attack persisted - escalating to time ban           |
| 10.0.0.22 | Uncertain  | 0.0000   | -     | Low      | Phase 1 | 1        | 10s      | 2026-09-07 14:00:37 | Attack Stopped                                      |
| 10.0.0.24 | Uncertain  | 0.0000   | -     | Low      | Phase 1 | 1        | 10s      | 2026-09-07 14:00:36 | Attack Stopped                                      |
| 10.0.0.21 | UDP Flood  | 0.6448   | 78.7% | High     | Phase 2 | 1        | 21s      | 2026-09-07 14:00:31 | Attack persisted - escalating to time ban           |
| 10.0.0.23 | Uncertain  | 0.0000   | -     | Low      | Phase 1 | 1        | 11s      | 2026-09-07 14:00:31 | Attack Stopped                                      |
| 10.0.0.20 | Uncertain  | 0.0000   | -     | Low      | Phase 1 | 1        | 11s      | 2026-09-07 14:00:27 | Attack Stopped                                      |
| 10.0.0.25 | Uncertain  | 0.0000   | -     | Low      | Phase 1 | 1        | 10s      | 2026-09-07 14:00:23 | Attack Stopped                                      |
| 10.0.0.17 | SYN Flood  | 0.6528   | 92.9% | Critical | Phase 2 | 1        | 5s       | 2026-09-07 14:00:14 | Attack persisted - escalating to time ban           |
| 10.0.0.18 | SYN Flood  | 0.6440   | 91.2% | Critical | Phase 2 | 1        | 5s       | 2026-09-07 14:00:14 | Attack persisted - escalating to time ban           |
| 10.0.0.19 | SYN Flood  | 0.6401   | 91.2% | Critical | Phase 2 | 1        | 5s       | 2026-09-07 14:00:14 | Attack persisted - escalating to time ban           |

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

\_\_\_\_\_  
SIGNATURE OVER PRINTED NAME  
**Network Administrator**

\_\_\_\_\_  
DATE